

Inference-Time Personalized Alignment with a Few User Preference Queries

Victor-Alexandru Pădurean¹, Parameswaran Kamalaruban², Nachiket Kotalwar³, Alkis Gotovos¹, Adish Singla¹

Max Planck Institute for Software Systems¹ Featurespace Innovation Lab, Visa² Carnegie Mellon University³



1. Motivation

- Population-level alignment methods (RLHF/DPO) struggle with capturing the diversity of highly individual preferences, from stylistic writing preferences to visual aesthetics.
- Inference-time methods offer more flexibility for personalization, but existing pipelines either elicit users' preferences in text or require many pairwise comparison queries.
- We study: *How can we align a generative model's response to a specific user on the fly, where the user's preferences need to be elicited with limited interaction?*

2. Contributions and Overview of our Approach

- We formalize inference-time personalized alignment with emphasis on practical, low-interaction settings where preferences are elicited with few queries.
- We develop USERALIGN, grounded in logistic bandits, achieving fast alignment by modeling user feedback as consistent and noise-free.
- We demonstrate few-query alignment in text and image domains with both simulated and real users, and release our implementation and datasets.

① Input: Generate concept art for a videogame hero players instantly connect with.

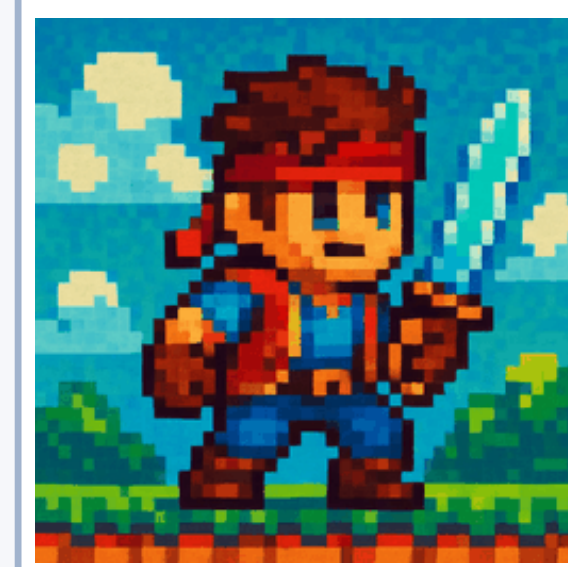
② System generates a diverse pool of responses



③ USERALIGN eliciting user preferences via pairwise comparisons



④ Output



Overview of our approach. Starting with a user question (Stage 1), the system generates a pool of responses (Stage 2). Then, USERALIGN iteratively collects user preferences (Stage 3) via pairwise comparisons to determine the most suitable response—user's preferred responses are **highlighted**. At the end, the final response is selected (Stage 4).

3. Problem Setup

- Task:** For a user prompt x , and a fixed pool of model responses $\mathcal{Y}_{\text{cand}}$, with a few pairwise comparisons, return $\hat{y} \in \mathcal{Y}_{\text{cand}}$ that matches the user's preference.
- Data:** Each user interaction logs (x, y, y', r) where $r = 1$ if y is preferred over y' , updating the dataset $\mathcal{D}$.
- Preference Model:** Responses are embedded via a feature map $\phi(x, y) \in \mathbb{R}^d$ and compared under the Bradley-Terry-Luce model. The best item in the pool is $y^* = \arg \max_{y \in \mathcal{Y}_{\text{cand}}} \langle \theta^*, \phi(x, y) \rangle$. We estimate θ^* on the fly with the data from $\mathcal{D}$.

4. Methodology: Theoretical Framework for UserAlign

System-User Interaction

- System uses the generative model π to generate a pool of responses $\mathcal{Y}_{\text{cand}}$ for x .
- while** stopping criteria not met **do**
 System selects a pair of responses $(y, y') \in \mathcal{Y}_{\text{cand}}^2$.
 System sets $r = 1$ if the user prefers y over y' , and $r = 0$ otherwise.
 System updates the preference dataset $\mathcal{D}$ with the preference tuple (x, y, y', r) .
- The system selects a final response $\hat{y} \in \mathcal{Y}_{\text{cand}}$ for the user.

Technical Details of the Algorithm

- Compute the norm-constrained unregularized MLE: $\hat{\theta}_t := \arg \min_{\|\theta\|_2 \leq S} \mathcal{L}_t(\theta)$
- Compute the loss-based confidence set:

$$\Theta_t := \left\{ \theta \in \mathbb{R}^d : \|\theta\|_2 \leq S \text{ and } \mathcal{L}_t(\theta) \leq \mathcal{L}_t(\hat{\theta}_t) + \beta_t \right\}$$
- Select the first response: $y_t^{(1)} \leftarrow \arg \max_{y \in \mathcal{Y}_{\text{cand}}} \langle \hat{\theta}_t, \phi(x, y) \rangle$
 Select the second response: $(y_t^{(2)}, \tilde{\theta}_t) \leftarrow \arg \max_{(y', \theta) \in \mathcal{Y}_{\text{cand}} \times \Theta_t} \langle \theta, \phi(x, y') - \phi(x, y_t^{(1)}) \rangle$
- Check the stopping condition $B(t) \leq \epsilon$, where $B(t) = \langle \tilde{\theta}_t, \phi(x, y_t^{(2)}) - \phi(x, y_t^{(1)}) \rangle$

Theoretical Analysis

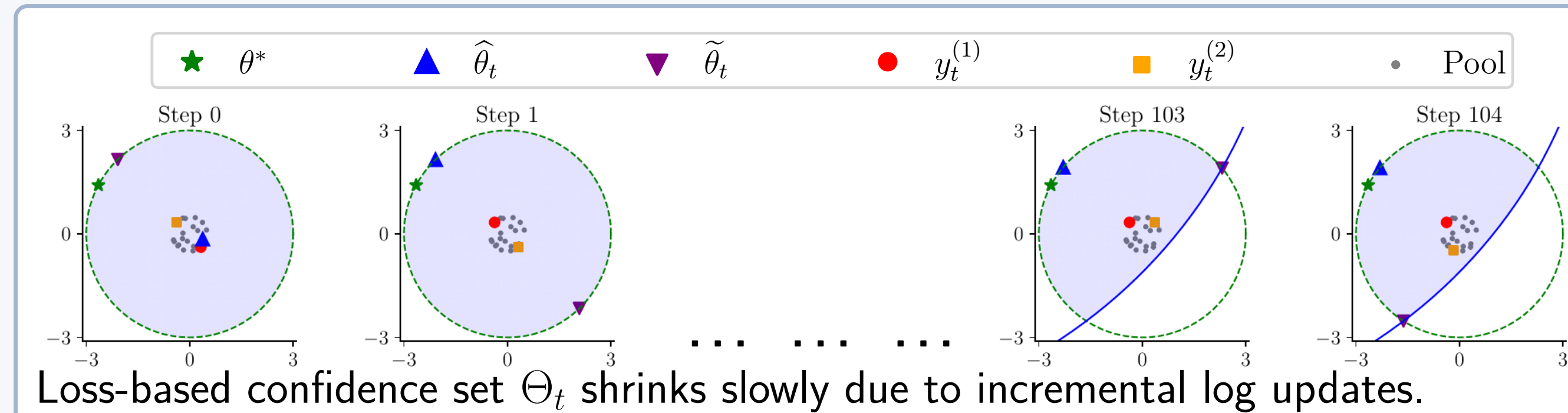
- Let τ be the stopping round. With probability at least $1 - \delta$, the final response is ϵ optimal: $\mathbb{P}[\langle \theta^*, \phi(x, y^*) - \phi(x, y_\tau^{(1)}) \rangle \leq \epsilon] \geq 1 - \delta$
- Let $K = |\mathcal{Y}_{\text{cand}}|$ and Ω be a constant depending on problem parameters. With probability at least $1 - \delta$, the number of rounds satisfies: $\tau \leq \mathcal{O}(\Omega K^2 \cdot \log(\Omega K^2))$

5. Methodology: Practical Confidence Set in UserAlign

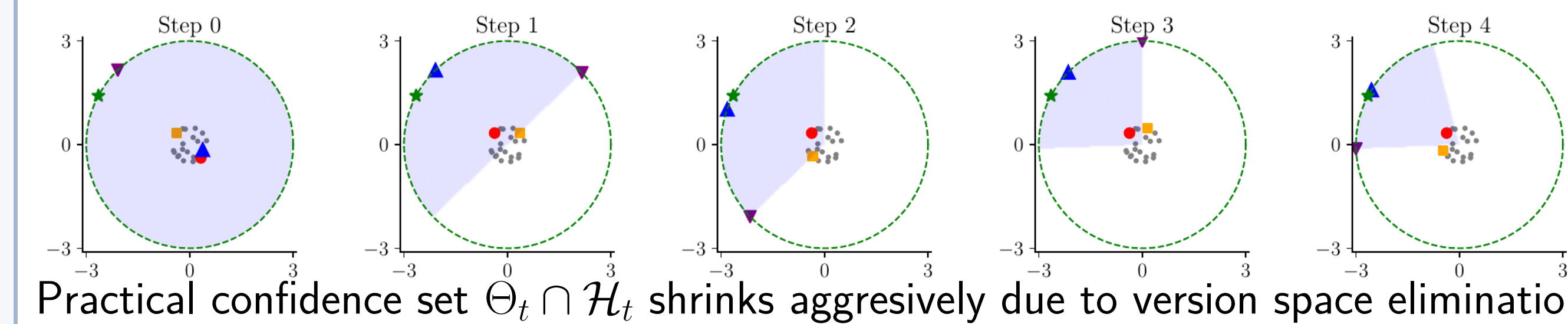
- The loss-based confidence set above shrinks slowly under incremental log-loss updates.

Key Idea: Treat the user's feedback as consistent and noise-free – leverage version space elimination via intersecting half-spaces to aggressively shrink the confidence set.

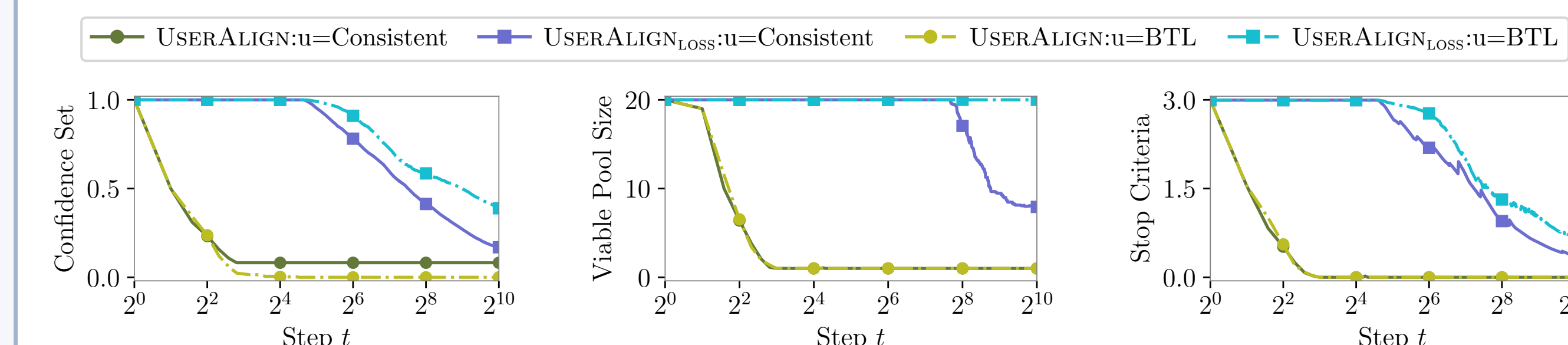
- Formally, $\mathcal{H}_t := \left\{ \theta \in \mathbb{R}^d : r_\tau \cdot \langle \theta, z_\tau \rangle - (1 - r_\tau) \cdot \langle \theta, z_\tau \rangle \geq 0 \text{ for all } \tau \in [t-1] \right\}$, where $z_\tau = \phi(x_\tau, y_\tau^{(1)}) - \phi(x_\tau, y_\tau^{(2)})$, and refine the loss-based set as $\Theta_t \leftarrow \Theta_t \cap \mathcal{H}_t$.



Loss-based confidence set Θ_t shrinks slowly due to incremental log updates.



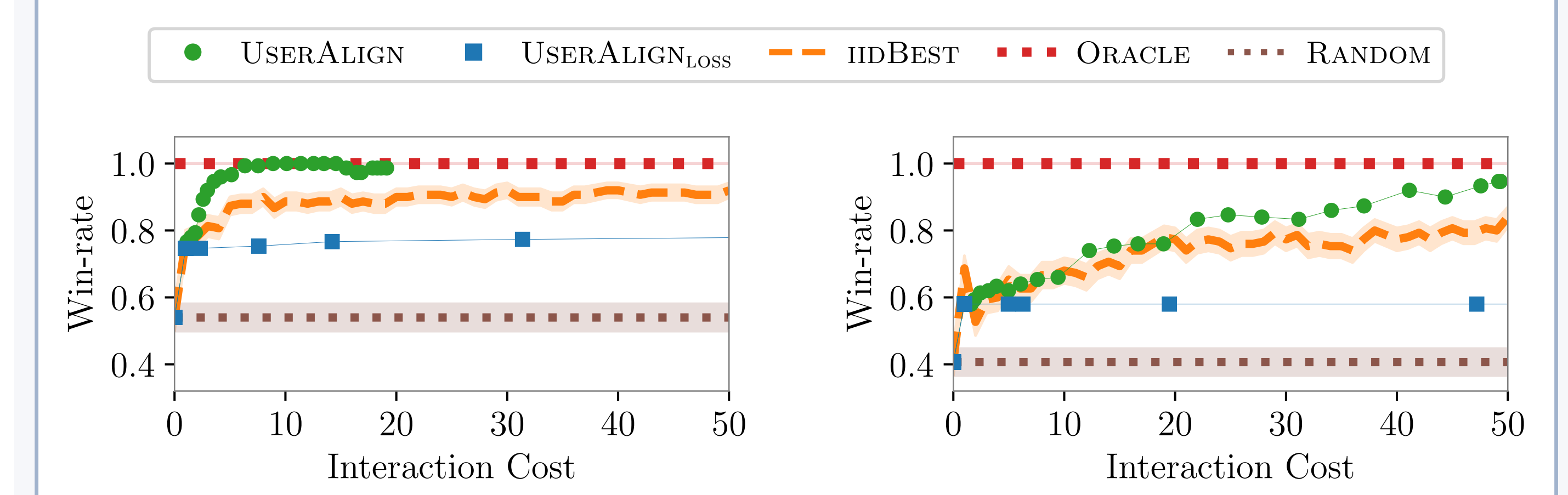
Practical confidence set $\Theta_t \cap \mathcal{H}_t$ shrinks aggressively due to version space elimination.



Normalized area of confidence set, viable response pool size, and stopping criteria over interaction steps. The practical confidence set shrinks significantly faster.

6. Experimental Evaluation

- Setup:** GPT-simulated users on a text-based food domain (food64d) and an image domain (visual1512d), with fixed, pre-generated candidate pools.
- Methods:** USERALIGN and USERALIGN_LOSS compared with IIDBEST (MLE on i.i.d. random pairs), ORACLE, and RANDOM. USERALIGN is based on the practical confidence set whereas USERALIGN_LOSS is based on the theoretical confidence set.
- Results:** USERALIGN achieves competitive win-rates at lower interaction costs, outperforming USERALIGN_LOSS and IIDBEST in high-dimensional preference spaces.



Win-rate vs. cost on text-based food personalization task (food64d).

Win-rate vs. cost on high-dimensional image personalization task (visual1512d).

7. Evaluation with Human Users

- Setup:** Human evaluation on Amazon Mechanical Turk, where participants choose responses according to personal preferences. Win-rates benchmark USERALIGN against IIDBEST at fixed query budgets, with $n = 120$ sessions per setting.
- Results:** USERALIGN consistently outperforms IIDBEST with larger gains at the later interaction steps.

Method	food64d win-rate (%)		visual1512d win-rate (%)	
	$t=5$	$t=10$	$t=10$	$t=20$
RANDOM	43.8 (3.2)	43.8 (3.2)	42.3 (3.2)	42.3 (3.2)
IIDBEST	73.3 (4.0)	76.7 (3.9)	78.3 (3.8)	79.2 (3.7)
USERALIGN	79.2 (3.7)	85.8 (3.2)	79.8 (3.7)	82.4 (3.5)

8. Conclusion

- We introduce USERALIGN, an inference-time method for efficiently aligning generative model responses to user preferences via sequential pairwise comparisons.
- Theoretical analysis and empirical evaluation show that USERALIGN provides speed-ups and cost reductions while maintaining strong alignment performance.
- Future work includes richer feedback (rankings/free-form), adaptive generation of candidates, and leveraging historical preferences to further reduce interaction cost.

Check the paper for full details on theoretical and experimental results:

